VIA PROTECTION DESIGN GUIDELINES 1.1

NCAB GROUP

Via protection is a crucial part of modern PCB production. Unfortunately, many people seem to struggle with choosing the right via protection type for their PCB, and we can't blame them. Even though the IPC provides a guideline regarding via protection (IPC 4761), it only describes the existing protection methods but doesn't provide any specification parameters, material choices, or design advice regarding the quality and reliability of the final PCBs. For this reason, we created this guideline to assist you in understanding and choosing the right via protection type for your PCB.

Definition

- The via protection describes a sub-process during PCB production that is executed after the copper plating but before the solder mask application.
- In this process, the plated-through holes are filled and/or covered with conductive or non-conductive material to improve the performance and reliability of the PCB.
- Solder resists and epoxy resins, as well as epoxy resins mixed with copper or silver particles, are used as filling materials. Depending on the filling material, a variety of silk screen process is used to achieve a sufficient filling of the via hole. Resing filling usually takes place under a vacuum.
- Via protection serves different purposes. When choosing the via protection type and filling material, the level of technology and the environment that the PCBA will be exposed to should be taken into consideration.

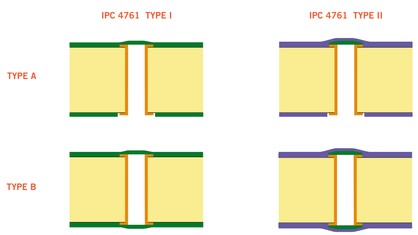
Purpose

- Protection against corrosion during plating process of the PCB and assembly process of the PCBA.
- Electrical and mechanical isolation of the via holes.
- Protection against solder mask creepage into the via holes.
- Protection against solder ball formation during wave or reflow soldering.
- Protection against unwanted solder paste loss and flux flow during the soldering process.
- Protection against the unwanted encapsulation of air or other substances under IC components.
- Improved thermal and/ or electrical performance of the vias by using special filler resins that contain metal particles.
- Improved vacuum performance by eliminating air leakage during assembly and ICT (In Circuit Test).
- Creation of a solderable via surface (via in pad - type VII only) often used in HDI applications.

General advice

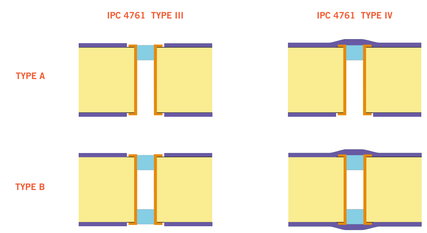
- The keywords “tenting, plugging, or filling,” without naming an exact type, make it difficult to understand our customers' needs, which results in additional time needed for clarification. Generally, we suggest via protection according to IPC 4761 Type VIb in most cases.
- If the vias in your Gerber data are not covered by solder resist and no via protection is specified in the procurement documentation, they will not be sealed. If there are vias with a diameter ≤ 0.35 mm on your PCB, we advise against this design, as unwanted solder resist residue can remain in the holes.
- If the vias in your design are covered with solder resist, we advise to fill them with solder resist in a separate process and later cover them on both sides during the regular solder resist process. This process is the most cost-effective way to achieve controlled and reliable via protection.

Via Tenting



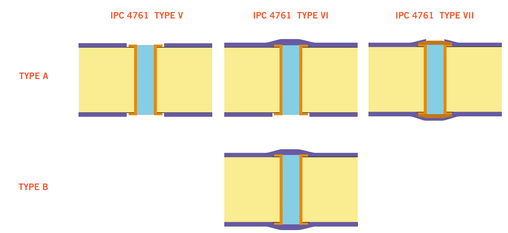
Diameter	n.a.	n.a.
Filling level	0%	0%
Material used	dryfilm (not available)	dryfilm (not available)
Solderability	no	no
Recommended by NCAB*	no (not available)	no (not available)

Via Plugging



Diameter	n.a.	n.a.
Filling level	< 70%	< 70%
Material used	soldermask ink (not recommended)	soldermask ink (not recommended)
Solderability	no	no
Recommended by NCAB*	no	no

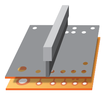
Via Filling

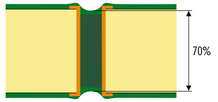


Diameter	≤ 0.5 mm	≤ 0.5 mm	≤ 0.5 mm
Filling level	>70%	≥ 70% (only applicable to type VI b when filling with solder mask) (NCAB unique)	100%
Material used	epoxy resin	solder mask / epoxy resin	epoxy resin
Solderability	no	no	yes
Recommended by NCAB*	yes (type VI preferred)	yes	yes

Your advantages with NCAB

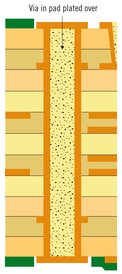
Contrary to the IPC 4761, we always specify the filling level of the vias to our factories. Without this, the filling level would be in an undefined state. NCAB factories aim for a filling level of 100%. A minimum filling level of 70% is required for Types V and VI, and 100% for Type VII. The via filling must always be performed in a separate screen-printing step and should not be combined with the regular solder mask application process.

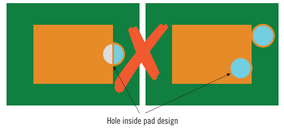




Design tips for PTH

- Always remove the solder mask openings for vias ≤ 0.35 that can to be covered. We advise to fill and cover them.
- If the whole annular ring needs to be covered in solder mask but the hole must remain open:
  - Soldermask opening 0.08 mm bigger than hole size but smaller than Pads,
  - Hole size ≥ 0.6 mm
  - For holes smaller than 0.6 mm, please contact your local NCAB FAE for review.
- Design tips to avoid unwanted Soldermask residue in via holes:
  - The most secure is hole size ≥ 0.6 mm to secure that holes will not be blocked or have Soldermask residues.
  - Hole size ≥ 0.40 mm secure no Soldermask in holes if making by E-spray (special control)
  - Hole Size ≥ 0.30 mm, no Soldermask in holes if done by low-press spray (special control)
- Holes inside pad, partly inside pad or too close to a pad is not suggested unless they are over-plated like IPC-4761 Type VII. (see illustration)
  - We recommend distance of ≥ 0.08 mm between such holes and edge of pad.
- Always protect vias within SMD pads according to IPC-4761 type VII.
- Epoxy filled vias should all be the same size if possible. Certainly no more than 0.15mm variation in size.





Hole sizes

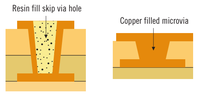
Compatible hole sizes (mm)

IPC 4761 Type V (resin non conductive)	0.20-0.50
IPC 4761 Type VI (solder mask)	0.20-0.50
IPC 4761 Type VI (resin non conductive)	0.20-0.50
IPC 4761 Type VI (resin electrically conductive)	0.20-0.50
IPC 4761 Type VI (resin thermally conductive)	0.20-0.50
IPC 4761 Type VII (plated/ capped)	0.20-0.50

For hole sizes outside this guideline, please consult your local NCAB technical contact to review specific projects case by case.

Design tips for micro vias

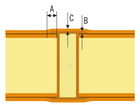
- Always copper fill microvias within SMD pads. It is more costly, but a better and more reliable option.
- Always resin fill skip via holes. Try to avoid skip via structures whenever possible, and choose staggered structures instead.
- For a higher reliability, we recommend that microvias should be 100 µm when copper filling is required.



Minimum track and gap for cap plating IPC-4761 TYPE VII PLUGGING / VIPPO / POFV

LOCATION	IPC DEMAND (Copper wrap & cap plating)	BASE COPPER (oz)	GENERAL TRACE	GENERAL GAP	MODERATE TRACE	MODERATE GAP**	ADVANCED TRACE	ADVANCED GAP**
Outer layer (PTH holes)	Class 2	1/3	N/A		0.10	0.10	0.09	0.10
		1/2	N/A		0.10	0.12	0.10	0.10
	Class 3	1/3	N/A		0.12	0.12	0.09	0.10
		1/2	N/A		0.12	0.12	0.10	0.10
Inner layer (Buried holes)*	Class 2	1/3	N/A		0.10	0.10	0.09	0.10
		1/2	N/A		0.10	0.12	0.10	0.10
	Class 3	1/3	N/A		0.10	0.10	0.09	0.10
		1/2	N/A		0.10	0.12	0.10	0.10

* Valid only for cap plated buried core - typically when a microvia is stacked on a buried hole.
** For design tighter than this, please consult your local NCAB technical contact to review specific projects case by case.
1 time wrap copper, need 0.02mm more spacing for compensation.



A Minimum copper wrap distance 25 um for class 1, class 2 and class 3
B Minimum copper wrap thickness 5 um for class 2 and class 3
C Minimum cap plating 12 um for class 3
C Minimum cap plating 5 um for class 2

Why we don’t recommend to use tenting or plugging

Both tenting and plugging only ensure that one or both sides of the via barrel are closed airtight. For tenting (IPC 4761 Types I & II), the entire via barrel surface remains unprotected, while for plugging (IPC 4761 Types III & IV), the majority of the via barrel surface is left unprotected.

Regarding IPC 4761, we recommend using Types V and above. Types V, VI, and VII ensure proper filling of the entire via hole, which is crucial for avoiding potential risks and ensuring consistent, reliable protection for all vias. This approach enables us to deliver high-quality, highly reliable PCBs that our customers can always trust.

Risk of using Tenting or Plugging
First of all, via tenting is an outdated technology and is no longer available in modern production facilities, so it’s off the table. Plugging (IPC 4761 Types III and IV), on the other hand, can be achieved but should be avoided. To understand why, it’s important to note that in most production facilities, the respective via protection is applied before the final surface finish (e.g., ENIG).

Applying a single-sided plug (IPC 4761 Types IIIa/IVa) in the via will prevent proper flow of the surface treatment chemistry, resulting in poor coverage and a severe risk of exposed copper in the holes. Exposed copper, of course, is prone to corrosion. Additionally, the cleaning solution at the end of the plating line will suffer from insufficient flow, so chemical residue may remain inside the hole after cleaning. These chemicals, of course, can accelerate potential corrosion.

So why not plug from both sides instead? Plugging the via barrel from two sides (IPC 4761 Types IIIb/IVb) introduces the risk of a "via blow-out" in your PCB. As the air trapped inside the via hole expands during the soldering process, the "plugs" inside the via openings can break due to excessive pressure within the via barrel.

For these reasons, methods that only partially cover or plug vias are neither recommended nor used by NCAB.

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